MGIS 355 Business Intelligence Project #3 Report

Project Group #4
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Introduction:

Majestic Media (M2) is a telecom company operating in the Northeast United States that has experienced significant growth in recent years. As market saturation has slowed new customer acquisition, the company's CEO has identified customer retention as a critical strategic priority. Customer churn, which is the rate at which customers discontinue service, represents a direct threat to revenue, particularly because acquiring new customers in a competitive market is substantially more costly than retaining existing ones.

The objective of this report is to develop and evaluate predictive models capable of identifying customers at risk of churning within a 12-month window. Using a dataset of 2,520 Majestic Media customers, three individual models and one ensemble model were constructed, compared, and evaluated across three dimensions: classification accuracy, Kaggle competition score, and monetary impact using a cost matrix. The final recommendation identifies the most effective model and provides actionable business insights to help M2 reduce its churn rate.

Data Preparation:

Dataset Overview:

Two datasets were provided for this analysis. The training dataset (Kaggle_churn_training.csv) contains 2,520 customer records with 13 variables including the target variable Churn. The test dataset (Kaggle_churn_test.csv) contains 630 customer records without the Churn label, used for Kaggle competition submission. The target variable is binary: 0 (no churn) and 1 (churn). Of the 2,520 training records, 2,123 customers (84.2%) did not churn and 397 customers (15.8%) churned, indicating a class imbalance that was addressed during modeling.

Exploratory Data Analysis:

Prior to modeling, exploratory analysis was conducted to understand data distributions, relationships between variables, and potential data quality issues. Key observations include:

- Class Imbalance: 84.2% of customers did not churn versus 15.8% who did. This imbalance was addressed using class_weight='balanced' in tree-based models and scale_pos_weight in XGBoost and LightGBM.
- Missing Values: Several columns contained missing values, notably Seconds_of_use, Frequency_of_use, Frequency_of_SMS, and Distinct_Called_Numbers, ranging from 1–12% missing. These were imputed using median imputation to preserve data distribution without introducing bias from extreme values.
- Complaint Status: Customers who filed complaints showed a substantially higher churn rate. This variable emerged as one of the strongest predictors across all models.

- Account Status: Non-active customers (Status = 2) churned at a dramatically higher rate than active customers (Status = 1), making Status the single most predictive feature in the dataset.
- Customer Value: Higher-value customers were somewhat less likely to churn, though the relationship was nonlinear. A log transformation was applied to normalize this skewed distribution.

Feature Engineering:
To improve model performance, 13 derived features were engineered from the original 12 predictors, resulting in a final feature set of 25 variables:

Feature	Formula	Rationale
avg_call_duration	Seconds_of_use / (Frequency_of_use + 1)	Captures typical engagement per call
failure_rate	Call_Failure / (Frequency_of_use + 1)	Normalizes failures by usage volume
sms_ratio	SMS / (calls + SMS + 1)	Communication channel preference
value_per_month	Customer_Value / (Subscription_Length + 1)	Monthly revenue contribution
distinct_ratio	Distinct_Called_Numbers / (Frequency_of_use + 1)	Diversity of call network
is_inactive	Status == 2	Binary flag for inactive accounts
has_complaint	Complains == 1	Binary flag for complaint history
log_customer_value	log(1 + Customer_Value)	Normalized customer value
log_seconds	log(1 + Seconds_of_use)	Normalized call volume
calls_per_second	Frequency_of_use / (Seconds_of_use + 1)	Call frequency density
is_contractual	Tariff_Plan == 2	Binary flag for contract type
high_charge	Charge_Amount >= 5	Flag for high billing tier
young_customer	Age_Group <= 2	Flag for younger age demographic

Table 1: Engineered Features

Data Partitioning:
The training dataset was used exclusively for model training and cross-validation. All models were evaluated using 3-fold stratified cross-validation to ensure each fold maintained the original 84:16 class ratio. The test dataset (630 records) was held out entirely and used only for generating the final Kaggle submission. No data leakage occurred between training and test sets.

Data Analysis (Model Building):
Three classification models were developed and evaluated. Each model was selected to represent a different algorithmic approach to the churn prediction problem. A fourth ensemble model was constructed by combining the three individual models. All models used the same 25-feature engineered dataset with median imputation applied.

Model 1 - Random Forest:
A random forest classifier was developed as the primary ensemble tree model. Random forests construct a large number of decision trees on bootstrapped subsets of data, each using a random subset of features at each split. The final prediction is determined by majority vote (for classification). This approach reduces overfitting and improves generalization compared to a single decision tree.

- **Algorithm:** Bagged ensemble of decision trees with random feature subsampling
- **Key Parameters:** n_estimators=500, max_features='sqrt', min_samples_leaf=2, class_weight='balanced', random_state=42, n_jobs=-1
- **Tuning Rationale:** 500 trees provide stable predictions while remaining computationally feasible. max_features='sqrt' is the standard recommendation for classification. class_weight='balanced' again adjusts for class imbalance.
- **Model Equation:** Churn = Majority_Vote($Tree_1$, $Tree_2$, …, $Tree_{500}$) where each tree = f(Status, Complains, Customer_Value, …)

Model 2 - XGBoost:
An XGBoost (Extreme Gradient Boosting) classifier was developed as a high-performance gradient boosting model. Unlike random forests which build trees in parallel, XGBoost builds trees sequentially, with each tree learning from the errors of the previous one. XGBoost includes built-in regularization (L1 and L2) to prevent overfitting and is widely regarded as a top-performing algorithm for tabular classification tasks.

- **Algorithm:** Gradient boosted decision trees with L1/L2 regularization
- **Key Parameters:** n_estimators=1000, max_depth=5, learning_rate=0.03, subsample=0.85, colsample_bytree=0.75, min_child_weight=3, gamma=0.1, reg_alpha=0.1, reg_lambda=1.0, scale_pos_weight=5.35

- **Tuning Rationale:** A low learning rate (0.03) with high n_estimators (1000) was used to learn slowly and precisely. scale_pos_weight=5.35 (ratio of negative to positive class) directly addresses class imbalance. subsample and colsample_bytree add stochastic regularization. reg_alpha and reg_lambda provide L1/L2 penalties to prevent overfitting.
- **Model Equation:** Churn = Σ_i f_i(Status, Complains, Customer_Value, is_inactive, has_complaint, …) where each f_i is a weak decision tree learner

Model 3 - LightGBM (Light Gradient Boosting Machine):
A LightGBM classifier was developed as the final model, utilizing a leaf-wise tree growth strategy rather than the level-wise approach used by standard Gradient Boosting. This allows for faster training speeds and higher accuracy on large datasets. LightGBM is highly efficient at handling the missing values (NA) present in our dataset without requiring manual imputation, and it is optimized for high-dimensional tabular data.

- **Algorithm:** Leaf-wise Gradient Boosted Decision TreesKey Parameters: n_estimators=500, learning_rate=0.05, num_leaves=31, scale_pos_weight=2.0, random_state=42
- **Tuning Rationale:** scale_pos_weight=2.0 was specifically chosen to align with the company's cost matrix, where a missed churner ($50) is exactly twice as costly as a caught churner ($25). A moderate learning_rate of 0.05 was used to ensure stable convergence, and num_leaves was kept at the default to maintain a balance between model complexity and generalization.
- **Model Equation:** Churn = $\sum_{i} f_i(\text{Status, Complains, Customer_Value, Call_Failure, ...})$ where each f_i is a leaf-wise decision tree focusing on minimizing the specific financial loss defined by the cost matrix.

<u>Results (Model Comparison):</u>

Confusion Matrices:
All models were evaluated using 3-fold stratified cross-validation on the training dataset. The confusion matrices below report the aggregated results across all three folds.

Model	TN	FP	FN	TP	Accuracy
Random Forest	2,061	62	79	318	94.40%
XGBoost	2,036	87	70	327	93.77%
LGBM	2,075	55	90	300	94.25%

Table 2: Confusion Matrices

The Random Forest model achieved the highest cross-validation accuracy at 94.40%, narrowly outperforming LGBM (94.25%) and XGBoost (93.77%). Notably, the Random Forest achieved the lowest false negative count (79 missed churners) among all models, which is particularly important given the cost matrix structure. XGBoost achieved the second-lowest false negative count (70), but its higher false positive count (87) reflects a more conservative prediction style. LGBM had the lowest false positive count (55), meaning it disturbed the fewest non-churning customers with unnecessary interventions.

Kaggle Submission Scores:
Each model's predictions were submitted to the Kaggle competition platform, which evaluates against a held-out test set using classification accuracy. The table below reports the Kaggle scores obtained for each model.

#	Model	Accuracy	Kaggle Score (Rank)	Username	Cost
1	Random Forest	94.40%	0.94285	joshuavannostrand	-$11,900
2	XGBoost	93.77%	0.93333	Joseph F Lombardo	-$11,675
3	LGBM	94.25%	0.9524	Joseph F Lombardo	-$11,250

Table 3: Kaggle Competition Results

Misclassification Cost Analysis:
To assess the monetary impact of each model, the cost matrix provided by Majestic Media's Customer Service department was applied to the cross-validation confusion matrices. The cost structure is:

	Predicted: 0 (No Churn)	Predicted: 1 (Churn)
Actual: 0 (No Churn)	$0	$0
Actual: 1 (Churn)	-$50	-$25

Table 4: Cost Matrix

Under this cost structure, each missed churner (false negative: actual churn, predicted no churn) costs -$50 because no intervention is made and the customer is lost. Each correctly predicted churner (true positive) costs -$25 because early intervention has a 50% chance of retention. False positives and true negatives carry no additional cost. Total model cost is therefore:

Total Cost = (FN × –$50) + (TP × –$25)

Model	FN (Missed Churners)	TP (Caught Churners)	FN Cost (@$50)	TP Cost (@$25)	Total Cost
Random Forest	79	318	-$3,950	-$7,950	-$11,900
XGBoost	70	327	-$3,500	-$8,175	-$11,675
LGBM	90	300	-$4,250	-$7,000	-$11,250

Table 5: Misclassification Cost by Model

XGBoost produced the lowest total cost (-$11,675) due to its low false negative count of 70 missed churners, meaning fewer customers were lost without any intervention. The LGBM model's low false positive count (55) is notable for customer satisfaction, as it avoided unnecessarily contacting loyal customers with retention offers. Random Forest produced a balanced cost profile (-$11,900) driven by its low overall false negative count, and achieved the highest raw accuracy among the three models.

Conclusion:

Model Recommendation:

Based on the three evaluation dimensions which are classification accuracy, Kaggle score, and monetary cost, our team recommends the Random Forest model as the primary model for deployment by Majestic Media, with the LightGBM (LGBM) model noted for achieving the highest Kaggle score on the held-out test set.

The Random Forest model achieved the highest cross-validation accuracy (94.40%) and the lowest false negative count (79 missed churners) among all three models, meaning fewer at-risk customers go undetected. While its total cost (-$11,900) is slightly higher than XGBoost (-$11,675), the difference is marginal and Random Forest's overall balance of precision, recall, and robustness makes it the most reliable choice for real-world deployment. The LGBM model's Kaggle score of 0.9524 demonstrates strong generalization to unseen data and its low false positive rate makes it particularly well-suited for scenarios where customer experience is a priority.

Business Recommendations:

Based on insights derived from the predictive models, the following recommendations are provided to Majestic Media to reduce customer churn:

Recommendation #1 - Prioritize Non-Active Customer Outreach:
Account Status (active vs. non-active) was the single most predictive variable across all models, with an XGBoost feature importance score of 30.6%. Customers with Status = 2 (non-active in the last month) churned at a dramatically higher rate. M2 should implement an automated early-warning system that flags accounts showing inactivity and triggers proactive outreach from the Customer Success team within the same month inactivity is detected. A reactivation incentive such as a service credit or promotional rates may be effective in re-engaging these customers before they formally terminate service.

Recommendation #2 - Resolve Complaints Rapidly and Follow Up:
Complaint history (Complains) was the second most important predictor (combined importance with has_complaint: ~26%). Customers who filed complaints were substantially more likely to churn. This finding indicates that M2's current complaint resolution process may be insufficient to restore customer satisfaction. The company should establish a dedicated churn-prevention protocol for complaint-filing customers: guaranteed response within 24 hours, escalation to senior support for unresolved issues, and a proactive follow-up call 7–14 days after resolution to confirm satisfaction. Measuring Net Promoter Score (NPS) post-resolution would help track improvement.

Recommendation #3 - Protect High-Value Customers with Loyalty Programs:
Customer_Value and its derived feature value_per_month were among the top predictors of churn risk. While higher-value customers were somewhat less likely to churn, the financial impact of losing a high-value customer is disproportionately large. M2 should implement a tiered loyalty program that provides premium customers with dedicated account managers, priority technical support, and exclusive retention offers. Identifying the top 20% of customers by Customer_Value and proactively engaging them with personalized retention offers would minimize the revenue impact of churn in this segment.

Recommendation #4 - Target Customers with High Call Failure Rates:
The engineered feature failure_rate (call failures normalized by usage volume) contributed meaningfully to model predictions. Customers experiencing frequent call failures relative to their usage are likely dissatisfied with service quality. M2 should use its network operations data to identify geographic or device-specific patterns in call failures and prioritize infrastructure improvements in high-failure areas. For individual customers flagged by the model as high churn-risk due to failure rate, a proactive service quality check and possible device upgrade offer could improve retention.

Final Summary:

In summary, Majestic Media's churn problem is most effectively addressed through a combination of predictive modeling and targeted operational interventions. The Random Forest model provides a reliable, interpretable, and accurate foundation for identifying at-risk customers. When deployed alongside the business recommendations above, particularly the proactive outreach for inactive accounts and complaint follow-up protocols, M2 can meaningfully reduce its churn rate, improve customer lifetime value, and strengthen its competitive position in the Northeast telecom market.